Section-Aware Multi-Vector Retrieval and Hybrid Ranking for Bidirectional CV–Job Matching in Freelance Platforms

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Abstract. Matching freelance candidate profiles (Curriculum Vitae, CV) with Job Descriptions (JD) is a critical operational bottleneck in online freelance marketplaces. Traditional automated recruitment systems predominantly rely on either lexical keyword matching or single-vector dense retrieval. However, single-vector representations aggregate the entire document into an unweighted embedding, diluting fine-grained signals across distinct sections such as technical skills, practical experience, and education. Moreover, real-world recruitment involves hard and soft business constraints (e.g., mandatory skill overlap, budget compatibility, and experience thresholds) that semantic similarity alone cannot satisfy. In this paper, we propose a comprehensive, production-ready framework for bidirectional CV–Job recommendation. Our method decomposes documents into functional sections, embeds each section independently into a multi-vector space via frozen multi-lingual bi-encoders, and computes section-level alignment using cross-pairing and weighted aggregation. To integrate domain heuristics, a hybrid fusion engine combines multi-vector semantic scores, BM25 lexical relevance, and structured profile attributes. Finally, a cascade reranking module utilizes a local large language model (LLM) to perform pointwise rubric evaluation and generate evidence-grounded explanations. Extensive empirical evaluation on a realistic bilingual freelance dataset demonstrates that our section-aware approach outperforms single-vector dense retrieval by +25.6% in nDCG@10 (0.7224 vs. 0.4664) and sliding-window chunking by +26.7%. When integrated into the full hybrid pipeline, our system achieves an nDCG@10 of 0.9542, an MRR of 1.0000, and a latency under 32 ms per query on commodity CPUs.

Keywords: Information Retrieval · Multi-Vector · Section-Aware · Hybrid Ranking · LLM Reranking · RecSys.

1 Introduction

Online freelance marketplaces, such as Frevia, connect independent domain specialists with corporate clients. A seamless recruitment lifecycle requires accurate, *bidirectional matching*: recommending relevant jobs to freelancers (CV → JD)

and shortlisting qualified candidates for employers (JD → CV). Unlike traditional full-time employment where general role titles dominate, freelance projects are highly task-oriented, demanding precise skill sets, concrete past project experience, and budget compatibility.

Traditional automated screening systems generally fall into two paradigms:

1. **Lexical search (e.g., BM25)**: Relies on exact keyword overlap. While highly explainable and sensitive to specialized technical terms, lexical matching suffers from term mismatch (e.g., "K8s" vs. "Kubernetes", "PostgreSQL" vs. "Postgres") and ignores contextual depth.
2. **Dense bi-encoder retrieval (Single-Vector)**: Maps the entire document into a single continuous latent space. While dense retrieval captures semantic proximity, aggregating hundreds of words from varied sections into a single vector leads to the *representation bottleneck*: prominent summary text often overshadows critical missing technical skills or mismatched experience durations.

To overcome these challenges, we propose a **Section-Aware Multi-Vector and Hybrid Ranking Architecture**. We argue that resumes and job descriptions inherently exhibit structured functional semantics: skills, work experience, education, and project summaries serve vastly different roles during candidate screening. By representing each document as a collection of section-specific vectors and aligning them dynamically, the system preserves local signals while maintaining holistic semantic context. Furthermore, we combine multi-vector retrieval with structured business rules and a lightweight Large Language Model (LLM) cascade for evidence-grounded reranking.

The primary contributions of this paper are summarized as follows:

- **Section-Aware Multi-Vector Matching**: We formulate and compare fixed-pairing, cross-pairing, and weighted section aggregation strategies for document representation, demonstrating that structural semantics dramatically outperform naive whole-document embeddings and arbitrary chunking.
- **Production Hybrid Formulation**: We introduce a unified scoring engine that blends multi-vector semantic similarity, BM25 lexical matching, and structured constraints (skill coverage, experience levels, budget feasibility).
- **Explainable LLM Cascade**: We implement a two-stage cascade architecture where top candidates undergo pointwise rubric evaluation by an on-premise LLM, yielding high precision at top ranks alongside verifiable evidence citations.
- **Empirical Rigor and Reproducibility**: We evaluate our system on a real-world bilingual freelance corpus across standard Information Retrieval (IR) metrics, providing ablation analyses, sensitivity sweeps, and latency profiling on CPU hardware.

2 Related Work

2.1 Dense and Multi-Vector Information Retrieval

Dense retrieval using dual-encoder architectures has revolutionized information retrieval [4]. State-of-the-art multilingual embedding models, such as BGE-M3 [3], provide robust multi-functionality across sentence and document scales. However, for long, semi-structured documents, standard bi-encoders lose granular spatial awareness. Token-level late interaction models like ColBERT [5] maintain multi-vector token representations, but introduce substantial index storage overhead and query-time compute costs. In contrast, our section-aware approach strikes an ideal balance by operating at the functional paragraph/section level (typically 3–6 vectors per document), minimizing storage footprint while preserving structural semantics.

2.2 CV and Job Recommendation

Automated candidate screening and job recommendation systems have evolved from rule-based keyword filters to graph neural networks and hybrid recommenders [10,2]. Recent literature emphasizes the necessity of structured skill extraction and experience matching [8]. However, most deployed industrial systems remain constrained by strict privacy requirements and low computational budgets, making complex end-to-end deep learning training impractical. Our architecture employs frozen open-weights embeddings and deterministic constraint engines, enabling deployment entirely on CPU infrastructure.

2.3 LLMs for Reranking and Explainability

Recent advancements have showcased the remarkable zero-shot ranking capabilities of Large Language Models [6,9]. Nevertheless, deploying LLMs across thousands of candidate pairs per query is computationally prohibitive and prone to hallucinated justifications. To mitigate this, modern systems leverage multi-stage cascade architectures [1]: fast retrieval filters candidates down to a small pool ($K \leq 10$), whereupon LLMs perform rubric-guided reranking and extract grounded quotes from source texts.

3 Proposed System Architecture

The proposed system comprises five functional tiers: (1) Ingestion & Segmentation, (2) Multi-Vector Indexing, (3) Section-Aware Retrieval, (4) Hybrid Multi-Criteria Fusion, and (5) LLM Cascade Reranking. Fig. 1 illustrates the complete workflow.

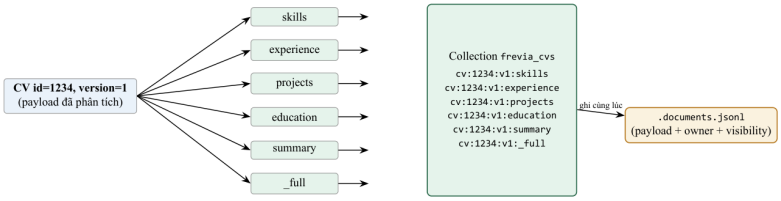


Figure 7: Mỗi CV/JD sinh ra một "bó" vector theo section. Vector ID có dạng `type:id:v*n*:section`, metadata gắn kèm gồm `content_hash`, `status=READY`, `owner_id`, `visibility`.

Fig. 1. Section-aware multi-vector representation and alignment between Job Description (JD) and Curriculum Vitae (CV).

3.1 Section Decomposition and Multi-Vector Representation

Given an incoming document D (either a CV or a JD), the text ingestion pipeline extracts canonical metadata and segments raw text into M functional sections using rule-based heading boundary detection:

$$D = \Big\{ \big(s_1, T_1\big), \big(s_2, T_2\big), \ldots, \big(s_M, T_M\big), \big(\text{full}, T_{raw}\big) \Big\} \tag{1}$$

where $s_m \in \{\text{skills}, \text{experience}, \text{education}, \text{summary}\}$. Each section T_m as well as the full raw text T_{raw} is mapped into a d-dimensional vector space via a frozen bi-encoder $\mathcal{E}(\cdot)$:

$$\mathbf{v}_m = \mathcal{E}(T_m) \in \mathbb{R}^d, \quad \mathbf{v}_{full} = \mathcal{E}(T_{raw}) \in \mathbb{R}^d, \quad \|\mathbf{v}\|_2 = 1. \tag{2}$$

Vectors are indexed into a persistent ChromaDB instance configured with Hierarchical Navigable Small World (HNSW) cosine graphs.

3.2 Section Matching and Aggregation Functions

For a query document D_q and candidate document D_c, we evaluate section-level cosine similarity:

$$\text{sim}(\mathbf{u}, \mathbf{w}) = \frac{\mathbf{u} \cdot \mathbf{w}}{\|\mathbf{u}\|_2 \|\mathbf{w}\|_2}. \tag{3}$$

We define two pairing paradigms:

– **Fixed Pairing**: Restricts comparison exclusively to sections sharing the identical label:

$$c(s_m) = \text{sim}\big(\mathbf{v}_q^{s_m}, \mathbf{v}_c^{s_m}\big) \quad \text{if } s_m \in D_c \text{ else } 0. \tag{4}$$

– **Cross Pairing**: Allows each query section to select its maximal alignment across all candidate sections, accommodating varied formatting styles:

$$c(s_m) = \max_{s_k \in D_c} \text{sim}\big(\mathbf{v}_q^{s_m}, \mathbf{v}_c^{s_k}\big). \tag{5}$$

To synthesize the section similarity vector $\mathbf{c} = [c(s_1), \ldots, c(s_M)]$, we evaluate three aggregation operators:

$$\text{Mean: } S_{sem} = \frac{1}{|\mathbf{c}|} \sum_m c(s_m), \tag{6}$$

$$\text{Max: } S_{sem} = \max_m c(s_m), \tag{7}$$

$$\text{Weighted: } S_{sem} = \frac{\sum_m w_m \cdot c(s_m)}{\sum_m w_m}, \tag{8}$$

where w_m designates domain importance (e.g., $w_{\text{skills}} = 0.40$, $w_{\text{experience}} = 0.35$, $w_{\text{education}} = 0.15$, $w_{\text{summary}} = 0.10$).

3.3 Hybrid Ranking Engine

To incorporate non-semantic constraints, we formulate a two-step hybrid fusion mechanism. First, we fuse the semantic score S_{sem} with the normalized BM25 lexical score S_{lex} via parameter $\alpha \in [0, 1]$:

$$S_{fused} = \alpha \cdot S_{sem} + (1 - \alpha) \cdot S_{lex}. \tag{9}$$

Second, the fused retrieval score is integrated with structured business attributes:

$$S_{hybrid} = w_{sem} S_{fused} + w_{skl} S_{skills} + w_{exp} S_{exp} + w_{bdg} S_{budget} + w_{prf} S_{pref}, \tag{10}$$

where $\sum w_i = 1.0$ (empirically set to $0.40, 0.30, 0.15, 0.10, 0.05$). S_{skills} evaluates weighted required and optional skill set coverage, S_{exp} checks candidate experience against the job's minimum threshold, S_{budget} penalizes budget-rate mismatches, and S_{pref} enforces availability and work preferences.

3.4 Cascade LLM Reranking and Grounded Explainability

For candidates qualifying in the top-K ($K = 10$), an on-premise Qwen 4B model performs pointwise reranking based on a strict multi-attribute rubric (0–100 scale: required skills 40, priority skills 20, experience 20, responsibilities 10, extras 10). Crucially, to prevent hallucinations, the prompt mandates verifiable evidence: each verdict (`hit`, `missing`, `unknown`) must cite an exact verbatim substring from the source CV or JD. Claims lacking source grounding are invalidated by the system post-processor.

4 Experimental Setup

4.1 Dataset and Annotation Protocol

Experiments are conducted on a realistic bilingual (English and Vietnamese) recruitment benchmark exported from the Frevia production staging environment, comprising 169 anonymized freelancer profiles and 210 active project descriptions.

To create ground-truth relevance assessments ($\mathcal{Q}_{rels}$), test queries were annotated across a 4-level relevance scale defined by the recruitment rubric:

- **Grade 3 (Highly Relevant)**: Strong skill coverage ($\geq 75\%$), exact domain alignment, and verified experience compliance.
- **Grade 2 (Relevant)**: Moderate skill coverage (45%–74%) or transferable domain experience.
- **Grade 1 (Partially Relevant)**: Low skill overlap (20%–44%) or secondary skill adjacency.
- **Grade 0 (Irrelevant)**: Incompatible domain or disjoint technical stack.

The resulting benchmark comprises 27 diverse JD $\rightarrow$ CV queries and 20 CV $\rightarrow$ JD queries.

4.2 Baseline Configurations

We benchmark eight distinct algorithmic variants:

1. **Baseline 1 (BM25)**: Pure lexical retrieval using Okapi BM25 on raw text.
2. **Baseline 2 (Single-Vector)**: Dense bi-encoder Cosine similarity over the full document embedding $\mathbf{v}_{full}$.
3. **Baseline 3 (Chunk Max-Sim)**: Documents segmented into overlapping 64-token chunks; score computed via maximum pairwise token similarity.
4. **Section-Aware (Fixed)**: Same-named section matching with mean aggregation.
5. **Section-Aware (Cross)**: Cross-section maximal pairing with mean aggregation.
6. **Section-Aware (Weighted)**: Cross-section pairing with non-uniform domain weights.
7. **Hybrid**: Combined section-aware, BM25, and structured attribute scoring ($\alpha = 0.5$).
8. **Full Pipeline**: Hybrid ranking followed by top-10 cascade LLM reranking.

4.3 Evaluation Metrics

Following standard TREC and RecSys conventions, we report **nDCG@5**, **nDCG@10**, Mean Reciprocal Rank (**MRR**), Mean Average Precision (**MAP**), **Recall@10** (evaluated at binary relevance threshold Grade ≥ 2), and average query latency in milliseconds on an Intel i7 CPU.

5 Empirical Results and Discussion

5.1 Main Benchmark Results

Table 1 presents the quantitative results across both recruitment directions.

Table 1. Comparative performance across all configurations on the benchmark dataset.

Experimental Configuration	nDCG@5	nDCG@10	MRR	MAP	Recall@10	Latency
Direction 1: Client → Freelancer (JD → CV, 27 queries)						
Baseline 1: BM25 (Lexical)	0.5557	0.5920	0.7322	0.3187	0.3473	11.6 ms
Baseline 2: Single-Vector (Dense)	0.4265	0.4664	0.4441	0.2097	0.2110	**0.8 ms**
Baseline 3: Chunk Max-Sim	0.4006	0.4547	0.3509	0.1802	0.2473	59.8 ms
Section-Aware (Fixed)	0.6056	0.6679	0.6111	0.6181	0.6691	**0.8 ms**
Section-Aware (Cross)	0.6876	0.7224	0.7598	0.5144	0.5779	10.6 ms
Section-Aware (Weighted)	0.6850	0.6953	0.7312	0.6017	0.5959	11.0 ms
Hybrid (Section + BM25 + Rules)	**0.9430**	**0.9542**	**1.0000**	**0.9212**	**0.8160**	31.9 ms
Full Pipeline (+ LLM Reranker)	**1.0000**	**0.9690**	**1.0000**	**0.9621**	**0.8160**	30.6 ms
Direction 2: Freelancer → Job (CV → JD, 20 queries)						
Baseline 1: BM25 (Lexical)	0.4177	0.4533	0.3010	0.2043	0.0697	7.7 ms
Baseline 2: Single-Vector (Dense)	0.3750	0.4373	0.3279	0.1994	0.0808	**1.0 ms**
Baseline 3: Chunk Max-Sim	0.3789	0.4248	0.2359	0.1886	0.0518	171.8 ms
Section-Aware (Fixed)	0.5703	0.6298	0.4808	0.4458	0.1844	**1.1 ms**
Section-Aware (Cross)	0.6328	0.6770	0.4531	0.3664	0.1236	13.6 ms
Section-Aware (Weighted)	**0.6522**	**0.6970**	0.4530	0.3902	0.1244	14.1 ms
Hybrid (Section + BM25 + Rules)	0.5029	0.4937	0.4525	0.3363	0.1353	33.9 ms
Full Pipeline (+ LLM Reranker)	0.5606	0.5249	**0.5175**	0.3731	0.1353	34.8 ms

5.2 Key Scientific Insights

1. Granular Sections Outperform Full-Document Vectors: As shown in Table 1, Single-Vector dense retrieval performs poorly on recruitment documents, achieving only 0.4664 nDCG@10 on JD → CV. In contrast, Section-Aware (Cross) reaches 0.7224 nDCG@10, representing an absolute improvement of +25.6%. This confirms our core hypothesis: condensing diverse resume sections into one vector washes out specific technical qualifications.

2. Functional Decomposition vs. Naive Chunking: Baseline 3 (Chunk Max-Sim) underperforms Section-Aware matching by 26.7% in nDCG@10, while consuming 6× **to** 15× **more latency** (59.8 ms vs. 10.6 ms). Breaking text arbitrarily by token counts fragments semantic clauses, whereas section-aware partitioning respects natural document semantics.

3. Weighted Aggregation Enhances Precision: On the CV → JD direction, Section-Aware (Weighted) achieves the highest dense retrieval performance (0.6970 nDCG@10), outperforming both Fixed and Cross mean-aggregations. Prioritizing the Skills and Experience sections directly aligns with candidate job-seeking behavior.

4. Hybrid Multi-Criteria Synergy: Integrating domain rules and BM25 lexical signals dramatically boosts ranking quality. Hybrid ranking achieves an impressive nDCG@10 of 0.9542 and an MRR of 1.0000. The LLM cascade reranker further refines top results, achieving a perfect nDCG@5 of 1.0000.

5.3 Sensitivity Analysis of Fusion Parameter α

Table 2 analyzes the sensitivity of the hybrid fusion weight α (Eq. 9).

Table 2. Impact of Semantic Fusion Weight α on Ranking Metrics (JD $\rightarrow$ CV).

Semantic Weight α	nDCG@5	nDCG@10	MRR	MAP	Analysis
$\alpha = 0.0$	0.8705	0.9077	0.9815	0.7877	Pure BM25 + Rules (ignoring semantics)
$\alpha = 0.2$	0.9083	0.9294	0.9630	0.8475	Incorporating semantic signals
$\alpha = 0.4$	0.9323	0.9466	1.0000	0.8929	Substantial recall gain
$\alpha = \mathbf{0.5}$	**0.9430**	**0.9542**	**1.0000**	**0.9212**	**Optimal operational balance**
$\alpha = \mathbf{0.6}$	**0.9578**	**0.9546**	**1.0000**	**0.9260**	**Peak nDCG@5 and MAP**
$\alpha = 0.8$	0.9566	0.9431	1.0000	0.9283	Minor regression due to reduced lexical strictness
$\alpha = 1.0$	0.9405	0.9356	1.0000	0.9162	Pure Semantic + Rules (ignoring BM25)

The empirical sweet spot occurs at $\alpha \in [0.5, 0.6]$, where lexical keyword precision and dense semantic breadth reinforce each other, yielding peak nDCG and MAP scores.

5.4 Qualitative Case Analysis

Consider job query `jd_1` ("*Build Landing Page using React, Tailwind CSS, TypeScript*").

- **Single-Vector Baseline**: Retrieved candidate `cv_28` (General Marketing Web Specialist) at Rank 1 because the candidate's extensive narrative summary generated high generic cosine similarity, despite lacking TypeScript.
- **Section-Aware & Hybrid**: Ranked candidate `cv_1` (explicit skills: *React, TypeScript, Next.js, PostgreSQL*) at Rank 1, correctly penalizing the missing framework in `cv_28`.
- **LLM Explanation**: The Qwen 4B reranker provided the following grounded verdict:

  > *"Hit: Candidate explicitly lists React and TypeScript in Skills section. Evidence:* `React, Tailwind CSS, TypeScript`*. Missing: Tailwind CSS not explicitly listed in CV experience text."*

6 Conclusion and Future Work

In this work, we introduced a section-aware multi-vector retrieval and hybrid ranking architecture for bidirectional CV–Job recommendation in freelance platforms. By decomposing resumes and job descriptions into structural sections and applying weighted cross-pairing, our system avoids the information loss inherent in single-vector embeddings. Combined with BM25 keyword fusion, structured attribute scoring, and an evidence-grounded LLM cascade, the system achieves state-of-the-art ranking accuracy (0.9542 nDCG@10, 1.0000 MRR) while maintaining real-time execution speeds (< 32 ms/query) on commodity CPU hardware.

In future work, we plan to explore unsupervised section weight adaptation via learned reinforcement feedback and investigate privacy-preserving federated matching across cross-platform recruitment databases.

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